GCP-GCVE Network Improvement Guide

Consolidated Greenfield Deployment and Brownfield Audit Playbook

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Date: 27 April 2026

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1. Introduction

This guide consolidates best practices for deploying and auditing Google Cloud VMware Engine (GCVE) connectivity with Google Compute Engine (GCE). It covers both Greenfield (new setup) and Brownfield (existing infra audit) scenarios.

2. Greenfield Deployment Checklist

Step	Category	Action Item	Purpose / Policy
1	Foundation	Deploy Shared VPC	Centralizes network control; separates roles.
2	IPAM	Reserve Non-Overlapping CIDRs	Prevents clashes between GCVE, GCE, and c
3	Connectivity	Establish Private Service Access (PSA)	Links GCVE Service Producer VPC to Shared
4	Routing	Enable Global Dynamic Routing	Propagates BGP routes across regions.
5	DNS	Configure Cloud DNS Peering	Ensures GCE <-> GCVE hostname resolution
6	GCE Security	Service Account-based FW Rules	Identity-based Zero Trust firewalling.
7	GCVE Security	NSX-T Distributed Firewall	Micro-segmentation to stop lateral movement.
8	Governance	Enable VPC Service Controls	Prevents data exfiltration to unauthorized proje

3. Brownfield Audit Framework

Audit Area	Current Gap	Improvement
Topology	Mesh Peering	Migrate to NCC Hub-and-Spoke
Internet Access	VMs with External IPs	Use Cloud NAT + IAP
Security	Allow-All FW Rules	Hierarchical FW Policies
Traffic Flow	Unencrypted traffic	Internal HTTPS LB + mTLS
Visibility	No traffic logs	Enable VPC Flow Logs + Insights
Hybrid Path	Single VPN Tunnel	HA VPN or Interconnect
Redundancy	Single Region	Global LB + Multi-Region GCVE

4. Implementation Steps

1. Discovery: Map all CIDRs (on-prem, GCE, other clouds).
2. Org Policy: Skip default VPC creation, restrict public IPs.
3. Peering Setup: Establish Shared VPC <-> GCVE peering.
4. Route Exchange: Enable Import/Export Custom Routes.
5. Validation: Run Connectivity Tests in Network Intelligence Center.

5. gcloud Command Reference

```
# Shared VPC
gcloud compute shared-vpc enable <HOST_PROJECT_ID>

# PSA
gcloud compute addresses create gcve-psa-range \
  --global --prefix-length=22 --purpose=VPC_PEERING \
  --network=<SHARED_VPC_NAME>

# VPC Peering
gcloud compute networks peerings create gcve-peering \
  --network=<SHARED_VPC_NAME> \
  --peer-project=<GCVE_PROJECT_ID> \
  --peer-network=gcve-network \
  --export-custom-routes --import-custom-routes

# Firewall Rules
gcloud compute firewall-rules create allow-gce-to-gcve \
  --network=<SHARED_VPC_NAME> \
  --action=ALLOW --rules=tcp:443 \
  --source-service-accounts=<SERVICE_ACCOUNT_EMAIL>
```

6. Architecture Diagram

Insert architecture diagram here (GCP & GCVE Hybrid Network Architecture).

7. References

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https://cloud.google.com/resource-manager/docs/organization-policy/constraints

https://cloud.google.com/vpc/docs/private-services-access

https://cloud.google.com/network-connectivity/docs/router

https://cloud.google.com/dns/docs/overview#peering

https://cloud.google.com/architecture/best-practices-vpc-design

https://cloud.google.com/vmware-engine/docs/best-practices-security

https://cloud.google.com/nat/docs/overview

https://cloud.google.com/iap/docs/using-tcp-forwarding

https://cloud.google.com/load-balancing/docs/https/internal-overview

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https://cloud.google.com/vmware-engine/docs/best-practices-networking